

PHYSICS 7B

UC BERKELEY

SPRING 2026

7B Discussion Problems, Week 7

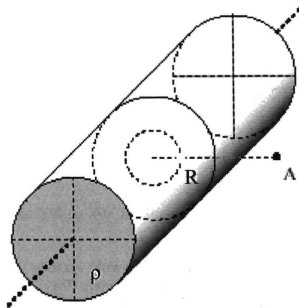
Topics

Continuous Charge Distributions, Gauss's Law, and the Electric Potential

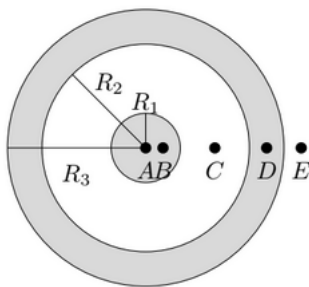
February 21, 2026

Week 7 Discussion Problems

- (Giancoli 22.5) The total electric flux from a cubical box 28.0 cm on a side is $1.84 \times 10^3 \text{ N}\cdot\text{m}^2/\text{C}$. What charge is enclosed by the box?
- A very long tube of radius R is full of charged stuff with uniform positive charge per unit volume ρ , as shown in the figure below.



- Sketch the electric field created by this charge distribution.
 - Find the electric field outside the tube, for $r > R$.
 - Find the electric field inside the tube, for $r < R$.
 - Is the electric field continuous at the surface of the cylinder? That is, are the values of $E_{\text{inside}}(r = R)$ and $E_{\text{outside}}(r = R)$ equal?
- A long cylinder of radius R_1 is surrounded by a concentric cylindrical tube with inner radius R_2 and outer radius R_3 . The inner cylinder is an insulator, and is uniformly charged with a charge density ρ . The outer cylindrical tube is a conductor with no net charge. What is the magnitude of the electric

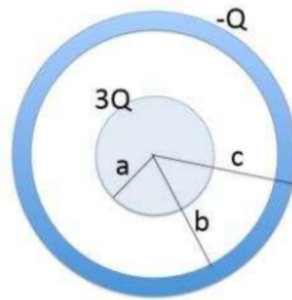


field at points A, B, C, D, and E (as a function of r , the distance to the cylinder's central axis)? What is the surface charge ρ_i on the inner surface of the conductor? What is the surface charge ρ_o on the outer surface of the conductor?

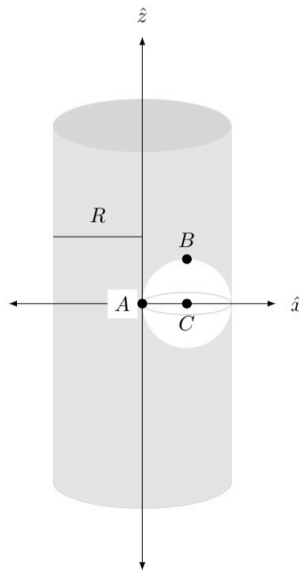
- A solid insulating sphere of radius a carries a net positive charge $+3Q$, uniformly distributed throughout its volume. Concentric with this sphere is a conducting spherical shell with inner radius b and outer radius c . The outer shell carries a charge of $-Q$ as shown in the figure below.

Find the electric field (magnitude and direction) at all locations:

- Within the sphere ($r < a$)
- Between the sphere and the shell ($a < r < b$)
- Inside the shell ($b < r < c$)
- Outside the shell ($r > c$)



- (e) What charges appear on the inner and outer surfaces of the shell?
5. The figure below shows an infinitely long insulating cylinder with uniform charge density ρ and a radius R . Inside the cylinder is an empty spherical cavity with radius $R/2$. What is the electric field (as a vector!) at points A, B, and C?



6. (Serway 23.47) An infinitely long cylinder of radius R has a volume charge density that varies with the radius as $\rho = \rho_0(a - \frac{r}{b})$ where ρ_0 , a , and b are positive constants and r is the distance from the axis of the cylinder. Use Gauss's law to determine the magnitude of the electric field at radial distances (s) $r < R$ and (b) $r > R$.
7. (*Potential from a nonuniform ring*) A non-uniformly charged ring of radius R carries a linear charge density $\lambda(\theta) = \lambda_0 \cos^2 \theta$ where $\lambda_0 > 0$ and θ is the angle with the horizontal axis. The point P is distance z above the center of the ring.
1. Draw a picture of the situation and label it.
 2. Calculate the electric potential at point P .

Source: Modified from Bordel, Midterm 2, Spring 2013.